



Enterobius vermicularis **(Pinworm)**

GIT module- Microbiology- LG4

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Introduction

- Cause disease known as enterobiasis, oxyuriasis or pinworm infection.
- Enterobiasis is highly prevalent throughout the world particularly in countries of temperate zone.
- *E. vermicularis* is considered to be **world's most common** parasite. The infection is common among children with age groups of 5-14 years, and is not associated with any specific socioeconomic level. It is usually prevalent in congested districts, schools, pre-schools, orphanages, mental institutions, day care centers.

Routes of transmission

Infection may be occurred by:

- Hand to mouth transmission. Hand contaminated with embryonated eggs through scratching the infected area or through handling contaminated articles (beddings, bed linens, night cloths....)
- Inhalation of air born eggs in dust.
- Rarely, retro-infection through the anus.

In households with several infected members, eggs may be found in dust of chairs, tables, pictures, floor and may be found in large numbers in the bedroom.

Morphology

Adult Worm

- The adults are short, white, fusiform worms with pointed ends, looking like bits of white thread. The mouth is surrounded by 3 wing-like cuticular expansions (cervical alae), which are transversely striated.
- The esophagus has a double-bulb structure, a feature unique to this worm

Female Worm

The female is 8–13 mm long and 0.3–0.5 mm thick.

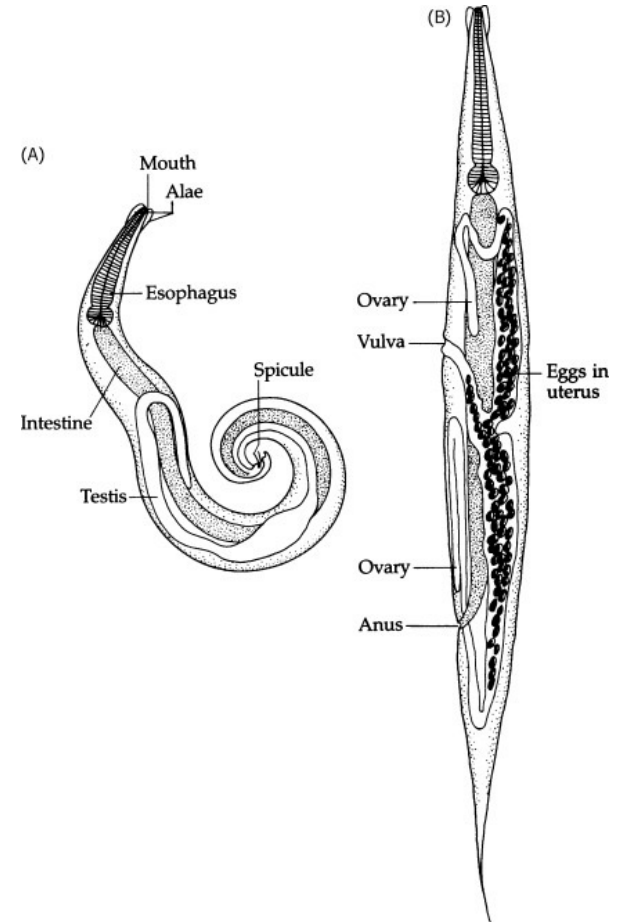
- Its posterior third is drawn into a thin pointed pin-like tail.
- In the gravid female, virtually the whole body is filled by the distended uteri carrying thousands of eggs.
- The worm is **oviparous**.
- Females survive for 5–12 weeks.

Male Worm

The male worm is 2–5 mm long and 0.1–0.2 mm thick. Its posterior end is tightly curved ventrally, and carries a prominent copulatory spicule. Males live for about 7–8 weeks.

Egg

- The egg is colorless and not bile-stained. It floats in saturated salt solution.
- It has a characteristic shape, being elongated ovoid, flattened on one side, and convex on the other (**planoconvex**), measuring 50–60 μm by 20–30 μm
- The egg shell is double layered and relatively thick, and transparent.
- The egg contains a coiled embryo, which is fully formed, but becomes infectious only 6 hours after being deposited on the skin. Under cool moist conditions, the egg remains viable for about 2 weeks.
- A single female worm lays 5,000–17,000 eggs.



Enterobius vermicularis (adult worms)



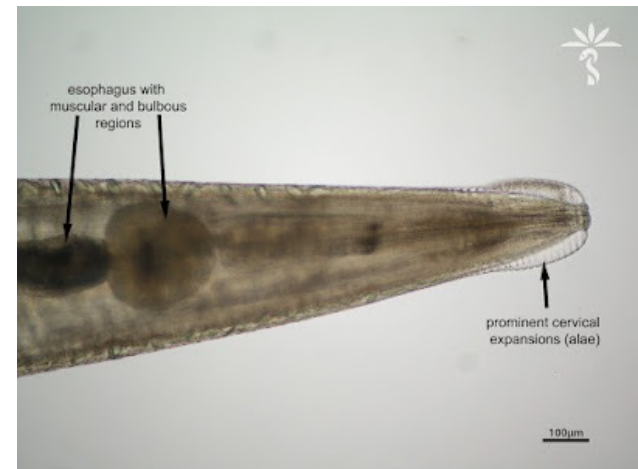
Enterobius vermicularis- Male



Enterobius vermicularis- Eggs



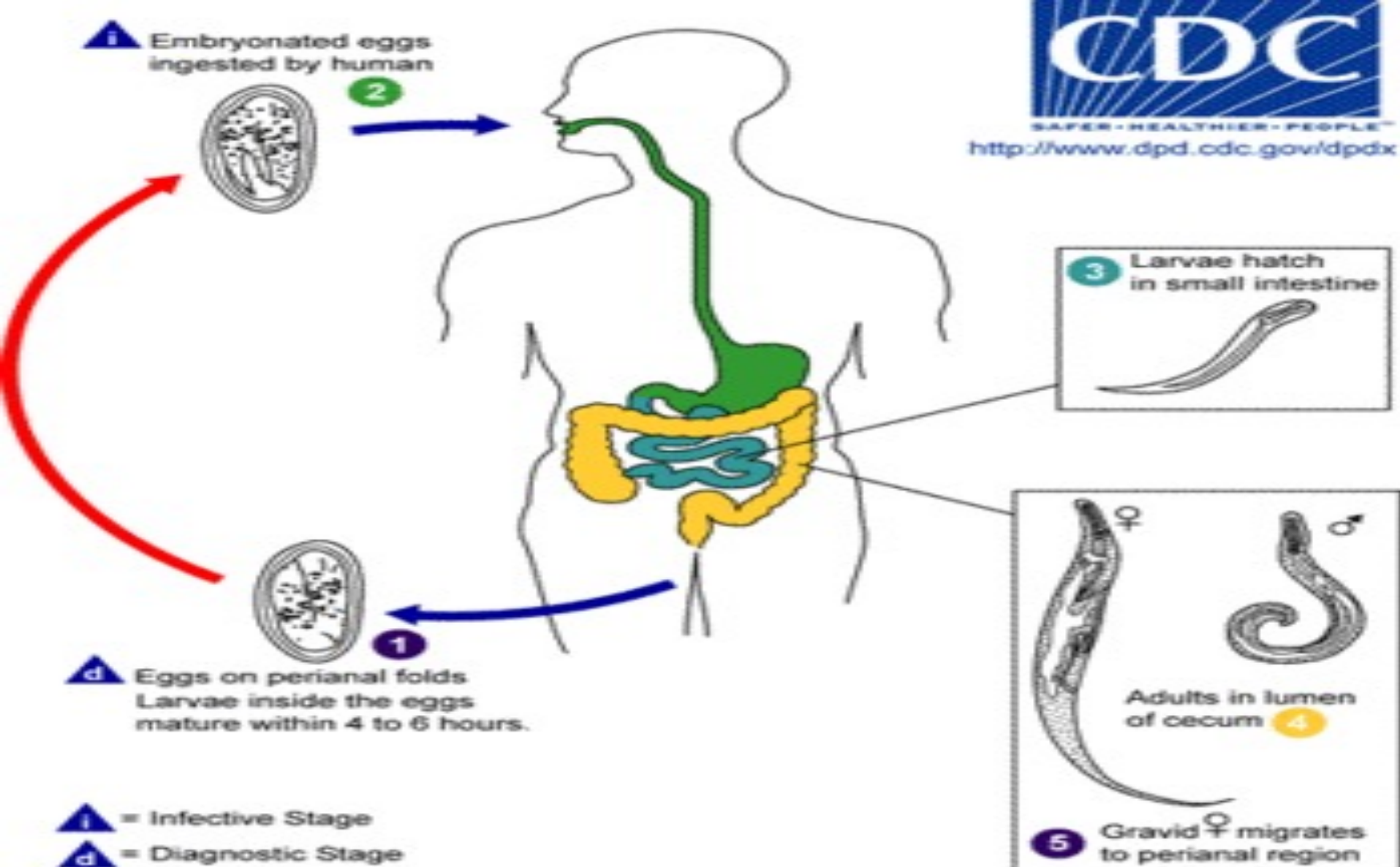
Enterobius vermicularis- Female



Enterobius vermicularis- Cervical alae

Life cycle

The adult worms inhabit the lumen of the cecum, appendix, and adjacent portion of ascending colon. At night gravid females migrate to the pre-anal region for oviposition (induced by low temperature and aerobic environment). The eggs embryonated within 6 hr. On ingestion, they hatch in the small intestine. The released larvae molt twice and develop into adults in about 36 - 53 days.



Life cycle of *Enterobius vermicularis*

Pathogenesis and symptomatology

Enterobiasis occurs mostly in children. It is more common in females than in males. About one-third of infections are asymptomatic.

- Migration of the gravid female to the pre-anal region and oviposition causing irritation and local pruritis of the perianal and perineal area (**pruritis ani**) which result in bleeding and bacterial infection.
- As the worm migrates out at night, it disturbs sleep, and **nocturnal enuresis** is sometimes seen.
- There is no evidence that eosinophilia and elevated level of serum IgE are associated with enterobiasis.

- Less frequently, the pathogenesis results from the intestinal activity of the parasite. The adults attach by their lips to the intestinal mucosa causing small ulcers which may lead to mild inflammation and bacterial infection.
- The worm crawling into the vulva and vagina causes irritation and a mucoid discharge. It may migrate up to the uterus, fallopian tubes and into the peritoneum. This may cause symptoms of **chronic salpingitis**, cervicitis, peritonitis, and recurrent urinary tract infections.
- The worm is sometimes found in surgically removed appendix and has been claimed to be responsible for **appendicitis**.

Diagnosis

The diagnosis is mainly based on the detection of the characteristic eggs in the pre-anal area using adhesive cellophan tapes. In heavy infection adult female can be seen at night in pre- anal region.

Treatment

Mebendazol 100 mg in a single dose and repeated two weeks later.

Prevention and control

- The finger nail should be cut short (in order to reduce hand - mouth transmission) and the hands should be washed thoroughly after using toilet and before having meals.
- Avoidance of scratching the infected area.
- Applying an ointment to reduce the eggs spread.
- Properly cleaning a potentially contaminated surfaces.
- All households should be treated to reduce and prevent further infections. (mass treatment)

References

- Paniker J.CK. 2018: Textbook of Medical Parasitology. 7thEd. Jaypee Brothers, ND. India.
- *Medical Parasitology, edited by Abhay R. Satoskar, Gary L. Simon, Peter J. Hotez and Moriya Tsuji. ©2009 Landes Bioscience*
- *Textbook of Medical parasitology, edited by Parija S.Ch. second edition © 2004, Chennai, New Delhi.*

Thanks a lot